CBSE Class 12 Pe — Physiology and Injuries in Sports

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

Q1. Match the types of fractures in List-I with their features in List-II: List-I (Fracture) (a) Transverse (b) Oblique (c) Greenstick (d) Comminuted List-II (Feature) i. Bone breaks diagonally ii. Bone is crushed into pieces iii. Straight break across the bone iv. Soft bone bends and cracks (A) a-iii, b-iv, c-ii, d-i (B) a-iii, b-i, c-iv, d-ii (C) a-i, b-ii, c-iii, d-iv (D) a-ii, b-iii, c-iv, d-i [PYQ 2024] | Confidence: Very High

Q2. Which one of the following factors does **NOT** determine flexibility? (A) Joint Structure (B) Previous Injury (C) Efficiency of Lungs (D) Age and Gender [PYQ 2024] | Confidence: Very High

Q3. Which of the following fitness components allows an athlete to give a better performance if they have more slow-twitch muscle fibers? (A) Speed (B) Strength (C) Endurance (D) Flexibility [PYQ 2024] | Confidence: Very High

Q4. Sprain is an injury of the: (A) Ligament (B) Muscle (C) Bone (D) Joint [PYQ 2024] | Confidence: Very High

Q5. In PRICE treatment for sports injuries, what does 'I' stand for? (A) Iceing (Ice) (B) Incline (C) Incision (D) Irritation [PYQ] | Confidence: Very High

Q6. Which of these is NOT a soft tissue injury? (A) Abrasion (B) Dislocation (C) Strain (D) Incision [PYQ 2025] | Confidence: Very High

Q7. What is cardiac output? (A) Blood pumped by the heart in one minute (B) Blood pumped by the heart in one beat (C) Blood pumped by the heart in one stroke (D) None of the above [PYQ] | Confidence: Very High

Q8. When the bone is broken into more than one piece or crushed at a single place, it is called a: (A) Comminuted fracture (B) Compound fracture (C) Simple fracture (D) Greenstick fracture [PYQ 2025] | Confidence: Very High

Q9. Lactic acid tolerance is a physiological factor that primarily determines: (A) Strength (B) Speed (C) Flexibility (D) Endurance [PYQ] | Confidence: High

Q10. The amount of air inhaled and exhaled in one normal breath without any extra effort is called: (A) Vital Capacity (B) Tidal Volume (C) Residual Volume (D) Minute Volume [PYQ 2024] | Confidence: Very High

Q11. Which of the following is NOT a short-term effect of exercise on the muscular system? (A) Accumulation of lactate (B) Increased blood supply (C) Muscular hypertrophy (D) Increased muscle temperature [PYQ 2025] | Confidence: High

Q12. Contusion is an injury classified under soft tissue injuries. It is also commonly known as a: (A) Bruise (B) Abrasion (C) Sprain (D) Tendon tear [PYQ] | Confidence: High

Q13. A fracture in a soft bone, where the bone bends and cracks but does not break completely into separate pieces, usually found in children, is known as a: (A) Compound fracture (B) Greenstick fracture (C) Spiral fracture (D) Transverse fracture [PYQ 2024] | Confidence: Very High

Q14. Muscular strength generally starts receding during the age of: (A) 25-30 years (B) 35-40 years (C) 45-50 years (D) 50-55 years [PYQ] | Confidence: Medium

Q15. The rubbing away of the upper layer of the epidermis due to friction with a rough surface is called: (A) Sprain (B) Laceration (C) Contusion (D) Abrasion [PYQ] | Confidence: Very High

Q16. Fast-twitch muscle fibers (Type II fibers) are white in color and are best suited for: (A) Long-distance running (B) Marathon (C) Explosive and fast movements like 100m sprint (D) Gymnastics [Practice] | Confidence: High

Q17. The amount of blood pumped by the left ventricle of the heart in one beat/stroke is called: (A) Cardiac Output (B) Stroke Volume (C) Blood Volume (D) Residual Volume [PYQ] | Confidence: Very High

Q18. A severe bone injury where the broken bone pierces through the skin, exposing it to the external environment, is called a: (A) Simple Fracture (B) Open/Compound Fracture (C) Comminuted Fracture (D) Impacted Fracture [Practice] | Confidence: High

Q19. Define 'Incision' as a sports injury. [Practice] | Confidence: Medium

Q20. Assertion (A): The risk of sports injuries can be significantly reduced by proper warming up before the activity. **Reason (R):** Warming up increases body temperature, elasticity of muscles, and blood circulation, preparing the body for intense physical exertion. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [Practice] | Confidence: Very High

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2-Mark

Q21. Differentiate between Stroke Volume and Cardiac Output. [Practice] | Confidence: Very High

Q22. Enlist any four types of bone fractures. [PYQ 2025] | Confidence: Very High

Q23. What does the acronym PRICE stand for in the immediate management of soft tissue injuries? [PYQ] | Confidence: Very High

Q24. State any two common soft tissue injuries and briefly explain one of them. [Practice] | Confidence: High

Q25. Mention any two physiological factors determining the speed of an athlete. [PYQ] | Confidence: High

Q26. Outline any two physiological differences between males and females in the context of sports performance. [PYQ] | Confidence: Medium

Q27. Define 'Vital Capacity' and 'Residual Volume' of the lungs. [Practice] | Confidence: High

Q28. Clearly differentiate between a 'Sprain' and a 'Strain'. [PYQ] | Confidence: Very High

Q29. List down any two long-term effects of regular physical exercise on the muscular system. [PYQ] | Confidence: High

Q30. What is a Greenstick fracture? Why is it predominantly seen in young children? [Practice] | Confidence: Very High

Q31. Highlight any two short-term effects of exercise on the cardiovascular system. [Practice] | Confidence: Medium

Q32. Define VO_2 max (Maximum Oxygen Uptake) and state its importance in endurance sports. [Practice] | Confidence: High

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3-Mark

Q33. Explain the PRICE procedure systematically as a first-aid treatment for a soft tissue injury. [PYQ] | Confidence: Very High

Q34. Describe the following physiological terms: (a) Residual Volume (b) Stroke Volume. [PYQ 2024] | Confidence: Very High

Q35. Explain how the composition of muscle fibers (slow-twitch and fast-twitch) affects the speed and endurance of an athlete. [PYQ] | Confidence: Very High

Q36. What is a joint Dislocation? How can a dislocation be managed immediately on the field? [PYQ] | Confidence: High

Q37. Elaborate on the long-term effects of regular exercise on the size and efficiency of the heart. [PYQ] | Confidence: High

Q38. Explain any three common physiological changes that occur in the human body due to the natural process of aging. [PYQ] | Confidence: Very High

Q39. Differentiate between 'Laceration' and 'Incision' with suitable sports examples. [Practice] | Confidence: High

Q40. Detail any three key physiological factors that determine the 'Endurance' capacity of an athlete. [Practice] | Confidence: Very High

Q41. What do you mean by an 'Impacted Fracture'? Briefly explain how it generally occurs. [Practice] | Confidence: Medium

Q42. State any three short-term physiological effects of rigorous exercise on the muscular system (e.g., lactate accumulation, increased blood supply). [PYQ 2025] | Confidence: High

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5-Mark (Long Answer/Case-Study)

Q43. Explain any four physiological factors determining strength. What are the long-term effects of regular exercise on the muscular system? [PYQ] | Confidence: Very High

Q44. Discuss in detail the physiological factors determining **Strength** and **Speed** in an athlete. [PYQ] | Confidence: Very High

Q45. Injuries are an inseparable part of sports. Define Soft Tissue Injuries. Classify and thoroughly explain any four types of soft tissue injuries (such as Sprain, Strain, Contusion, Laceration, or Abrasion) along with their immediate first-aid management. [Practice] | Confidence: Very High

Q46. Case Study: Rohan, a 16-year-old student, got injured after colliding with Sachin during a football match. He was having unbearable pain in his ankle, and it was swollen. The coach noticed a collection of blood outside the blood vessel on Sachin's forehead, too. He immediately gave first aid to both and rushed them to the hospital. (i) What type of injury did Rohan suffer in his ankle? (ii) Sachin had a _ **on his forehead. (iii) If Sachin suffered an abrasion during the injury, which of the following symptoms would he have? (A) Tear of ligament (B) Rubbing away of upper layer of epidermis (C) Tear of muscles (D) Instability of joints. (iv) A fracture in a soft bone, in which the bone bends, is known as a** _. [PYQ 2024] | Confidence: Very High

Q47. Case Study: Diya went to a sports training centre for the first time. Her coach informed her that participation in sports not only promotes physical growth but also has social and psychological benefits. He highlighted numerous physical benefits for muscles, heart, and respiratory systems. He advised her to continue daily practice to improve her health-related and skill-related fitness. (i) Which is NOT a long-term effect of exercises on the muscular system? (A) Hypertrophy of muscle (B) Increase in glycogen stored (C) Ligament and tendon strengthen (D) Accumulation of lactate (ii) What is cardiac output? (iii) Choose the correct statement related to tidal volume: (A) Amount of air inhaled and exhaled in one normal breath. (B) Amount of air inhaled in a forced breath. (C) Amount of blood pumped out by the heart in one stroke. (D) Amount of air exhaled forcefully. (iv) Lactic acid tolerance relates to ___ (Strength / Speed / Flexibility / Endurance). [PYQ 2025] | Confidence: Very High

Q48. Case Study: The Godavari school attended a CBSE Cluster Basketball Tournament. During the semi-final match, Varun, one of the players, fell down and was injured on the shoulder. He was immediately given first aid by the coach Mr. Rahul, who had knowledge of first aid. Warm-up sessions are essential for players to avoid any serious injuries during the match. (i) The breakage of bones due to a heavy impact is called a _***. (A) Fracture (B) Sprain (C) Contusion (D) Laceration (ii) Contusion is a soft-tissue injury also commonly known as a*** _. (A) Bruise (B) Abrasion (C) Bone tear (D) Tendon snap (iii) The first-aid given immediately to a sprain or strain injury is ___. (A) RICER (B) Heat massage (C) Applying muscle ointment immediately (D) Surgery (iv) What is the main objective of warming up before a match? [PYQ] | Confidence: High

Q49. Define 'Bone Fracture'. Comprehensively classify and explain the mechanics of any four types of bone fractures (e.g., Simple, Compound, Comminuted, Transverse, Greenstick, or Impacted fracture). [Practice] | Confidence: Very High

Q50. Regular training profoundly impacts human physiology. Discuss any five long-term physiological changes that occur specifically in the **Cardiorespiratory System** due to sustained physical training. [Practice] | Confidence: Very High

Answer Key

Q1. (B) a-iii, b-i, c-iv, d-ii **Q2.** (C) Efficiency of Lungs **Q3.** (C) Endurance **Q4.** (A) Ligament **Q5.** (A) Iceing (Ice) **Q6.** (B) Dislocation **Q7.** (A) Blood pumped by the heart in one minute **Q8.** (A) Comminuted fracture **Q9.** (D) Endurance **Q10.** (B) Tidal Volume **Q11.** (C) Muscular hypertrophy (This is a long-term effect) **Q12.** (A) Bruise **Q13.** (B) Greenstick fracture **Q14.** (B) 35-40 years **Q15.** (D) Abrasion **Q16.** (C) Explosive and fast movements like 100m sprint **Q17.** (B) Stroke Volume **Q18.** (B) Open/Compound Fracture **Q19.** An incision is a clean, sharp cut in the skin or tissue, usually caused by a sharp-edged object or sports equipment like a skate blade. **Q20.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q21.** Stroke Volume: Amount of blood pumped by the heart in one beat. Cardiac Output: Amount of blood pumped in one full minute ($HeartRate \times StrokeVolume$). **Q22.** 1. Transverse 2. Oblique 3. Greenstick 4. Comminuted. **Q23.** P-Protection, R-Rest, I-Ice, C-Compression, E-Elevation. **Q24.** Examples: Sprain, Contusion. A sprain is a severe stretching or tearing of ligaments at a joint. **Q25.** 1. Muscle fiber composition (more fast-twitch fibers). 2. Explosive strength / Neuromuscular coordination. **Q26.** 1. Males generally have higher bone density and muscle mass. 2. Males have larger heart sizes and higher VO_2 max compared to females. **Q27.** Vital Capacity: The maximum amount of air a person can exhale after a maximal inhalation. Residual Volume: The volume of air still remaining in the lungs after a forced expiration. **Q28.** Sprain is a ligament injury (bone to bone). Strain is a muscle or tendon injury (muscle to bone). **Q29.** 1. Muscular Hypertrophy (increase in muscle size). 2. Increase in the number and size of mitochondria. **Q30.** A greenstick fracture is an incomplete fracture where the bone bends and cracks. It is common in children because their bones are softer, more flexible, and contain more cartilage. **Q31.** 1. Increase in heart rate. 2. Increase in stroke volume and cardiac output. **Q32.** VO_2 max is the maximum volume of oxygen the body can consume and utilize during intense exercise. High VO_2 max indicates superior aerobic endurance. **Q33. Protection:** Protect the injured area from further harm. **Rest:** Stop activity immediately. **Ice:** Apply ice packs for 15-20 mins to reduce swelling. **Compression:** Wrap with an elastic bandage to minimize swelling. **Elevation:** Keep the injured part raised above heart level to decrease blood flow/swelling. **Q34.** (a) Residual Volume: The amount of air left in the lungs after a maximal, forceful exhalation. (b) Stroke Volume: The volume of blood pumped out by the left ventricle of the heart during one systolic contraction (one beat). **Q35.** Fast-twitch fibers (white fibers) contract quickly and forcefully, aiding in high-speed, explosive events like sprinting. Slow-twitch fibers (red fibers) contract slowly but are highly resistant to fatigue, essential for endurance events like marathons. **Q36.** Dislocation is the displacement of bones from their normal position in a joint. Management: Do not try to pop it back forcefully. Immobilize the joint using a sling or splint, apply ice, and seek immediate professional medical attention. **Q37.** Regular endurance training causes "Cardiac Hypertrophy" (enlargement and strengthening of the heart muscle, especially the left ventricle). This leads to a lower resting heart rate, increased stroke volume, and highly efficient blood pumping. **Q38.** 1. Decrease in bone density and muscle mass. 2. Decrease in maximum heart rate and lung capacity. 3. Slower reaction time and reduced joint flexibility. **Q39.** Laceration is an irregular, jagged tear of the skin due to a blunt force or rough impact (e.g., falling on gravel). Incision is a smooth, clean cut caused by a sharp object (e.g., a cut from an ice skate blade). **Q40.** 1. VO_2 Max (maximum oxygen uptake). 2. Lactic Acid Tolerance (ability to sustain work despite lactic acid buildup). 3. Muscle Fiber Type (higher percentage of slow-twitch red fibers). **Q41.** Impacted fracture happens when the broken ends of a bone are forcefully driven or jammed into each other. It generally occurs due to a high-impact fall or a direct heavy blow along the axis of the bone. **Q42.** 1. Accumulation of lactic acid causing fatigue. 2. Increased muscle temperature. 3. Increased blood supply and oxygen demand in the working muscles. **Q43.** Factors determining strength: 1. Muscle Cross-Sectional Area (size). 2. Muscle fiber composition (fast-twitch). 3. Age & Gender. 4. Intensity of nerve impulse. Long-term effects: Hypertrophy (increased size), strengthening of tendons/ligaments, increased mitochondria, delayed fatigue. **Q44. Strength factors:** Muscle cross-sectional area, Muscle fiber type (Type II), Body weight, Age and gender. **Speed factors:** Muscle fiber composition (dominance of fast-twitch fibers), Neuromuscular responses/coordination, Explosive strength, Flexibility, and ATP energy stores. **Q45.** Soft tissue injuries involve skin, muscles, tendons, and ligaments. 1. **Sprain:** Ligament tear (Use PRICE). 2. **Strain:** Muscle/tendon tear (Use PRICE). 3. **Contusion:** Bruise from blunt trauma, internal bleeding (Use ice). 4. **Abrasion:** Scraping of the skin (Wash, apply antiseptic). 5. **Laceration:** Irregular cut. First-aid generally follows the PRICE (Protect, Rest, Ice, Compress, Elevate) methodology. **Q46.** (i) Sprain (ligament injury in the ankle). (ii) Contusion (blood collected outside vessels). (iii) (B) Rubbing away of upper layer of epidermis. (iv) Greenstick fracture. **Q47.** (i) (D) Accumulation of lactate (it is a short-term effect). (ii) The amount of blood pumped by the heart in one minute. (iii) (A) Amount of air inhaled and exhaled in one normal breath. (iv) Endurance. **Q48.** (i) (A) Fracture. (ii) (A) Bruise. (iii) (A) RICER (Rest, Ice, Compression, Elevation, Referral). (iv) To increase body temperature, muscle elasticity, and prevent soft tissue injuries. **Q49.** A fracture is a break in the continuity of a bone. 1. **Simple Fracture:** Bone breaks but skin is intact. 2. **Compound/Open:** Broken bone pierces the skin. 3. **Comminuted:** Bone crushes into 3 or more pieces. 4. **Greenstick:** Bone bends and cracks (common in kids). 5. **Transverse:** Straight break across the bone shaft. **Q50.** Long-term effects on Cardiorespiratory system: 1. **Cardiac Hypertrophy:** Heart size and strength increase. 2. **Decreased Resting Heart Rate:** Heart pumps more efficiently. 3. **Increased Stroke Volume & Cardiac Output:** More blood/oxygen delivered per beat. 4. **Increased Lung Capacity & Vital Capacity:** Better oxygen intake. 5. **Faster Recovery Rate:** Heart rate returns to normal quicker post-exercise.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).